

Hilbert Spaces: Lecture Notes

Transcribed from Handwritten Notes

Lectures 17–20

1 Lecture 17: Inner Product Spaces and Norms

Lemma 1.1 (Cauchy-Schwarz Inequality). *Let X be an inner product space and let $x, y \in X$. Then:*

$$|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$$

Proof. (a) If either $x = 0$ or $y = 0$, the result is true (both sides are 0). (b) Assume neither x nor y is zero. Let $\alpha \in \mathbb{F}$. By the definition of the inner product, $\langle \alpha x + y, \alpha x + y \rangle \geq 0$. Expanding this:

$$\begin{aligned} 0 &\leq \langle \alpha x + y, \alpha x + y \rangle \\ &= |\alpha|^2 \langle x, x \rangle + \alpha \langle x, y \rangle + \bar{\alpha} \langle y, x \rangle + \langle y, y \rangle \end{aligned}$$

We choose a specific value for α to simplify the expression. Let $\alpha = -\frac{\overline{\langle x, y \rangle}}{\langle x, x \rangle}$. Substituting this back into the expansion:

$$\begin{aligned} 0 &\leq \frac{|\langle x, y \rangle|^2}{\langle x, x \rangle^2} \langle x, x \rangle - \frac{\overline{\langle x, y \rangle}}{\langle x, x \rangle} \langle x, y \rangle - \frac{\langle x, y \rangle}{\langle x, x \rangle} \overline{\langle x, y \rangle} + \langle y, y \rangle \\ 0 &\leq \frac{|\langle x, y \rangle|^2}{\langle x, x \rangle} - 2 \frac{|\langle x, y \rangle|^2}{\langle x, x \rangle} + \langle y, y \rangle \\ 0 &\leq -\frac{|\langle x, y \rangle|^2}{\langle x, x \rangle} + \langle y, y \rangle \end{aligned}$$

Multiplying by $\langle x, x \rangle$ (which is positive):

$$|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$$

□

1.1 Induced Norms

The function $\| \cdot \| : X \rightarrow \mathbb{R}$ defined by $\|x\| = \langle x, x \rangle^{1/2}$ is a norm on X .

Proof. (i) $\|x\| = \langle x, x \rangle^{1/2} \geq 0$.

(ii) $\|x\| = 0 \iff \langle x, x \rangle = 0 \iff x = 0$.

(iii) $\|\alpha x\| = \langle \alpha x, \alpha x \rangle^{1/2} = (|\alpha|^2 \langle x, x \rangle)^{1/2} = |\alpha| \|x\|$.

(iv) (Triangle Inequality):

$$\begin{aligned}
\|x + y\|^2 &= \langle x + y, x + y \rangle \\
&= \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 \\
&= \|x\|^2 + 2\operatorname{Re}\langle x, y \rangle + \|y\|^2 \\
&\leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \quad (\text{Since } \operatorname{Re}(z) \leq |z|) \\
&\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \quad (\text{By Cauchy-Schwarz}) \\
&= (\|x\| + \|y\|)^2
\end{aligned}$$

Taking square roots implies $\|x + y\| \leq \|x\| + \|y\|$. □

Remark 1.1. The norm $\|\cdot\|$ defined above is said to be **induced** by the inner product.

1.2 Polarization Identities

Lemma 1.2. Let X be an inner product space. Then for all $u, v, x, y \in X$:

(a) $\langle u + v, x + y \rangle - \langle u - v, x + y \rangle = 2\langle u, x + y \rangle$ (Wait, looking at the expansion below, the notes derive the polarization identity).

Correction based on notes derivation:

(b) $\langle u + v, x + y \rangle - \langle u - v, x + y \rangle = 2\langle v, x \rangle + 2\langle v, y \rangle$. (Actually, the notes establish simpler identities first).

Soln:

(a) $\langle u + v, x + y \rangle = \langle u, x \rangle + \langle u, y \rangle + \langle v, x \rangle + \langle v, y \rangle$. $\langle u - v, x + y \rangle = \langle u, x \rangle + \langle u, y \rangle - \langle v, x \rangle - \langle v, y \rangle$. Subtracting these gives:

$$\langle u + v, x + y \rangle - \langle u - v, x + y \rangle = 2\langle v, x \rangle + 2\langle v, y \rangle$$

(b) $\langle u + iv, x + iy \rangle - \langle u - iv, x - iy \rangle = \dots$ (Using similar algebraic expansion).

Theorem 1.3 (Parallelogram Law and Polarization Identity). Let X be an inner product space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$. Then for all $x, y \in X$:

(a) **The Parallelogram Rule:**

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

(b) If X is Real then:

$$4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2$$

(c) If X is Complex then:

$$4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2$$

Proof. Proof follows immediately from the expansion of inner products. □

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Example 2.1. The standard norm (i.e., Sup norm) on the space $C[0, 1]$ is **not** induced by an inner product.

Solution. Consider the functions $f, g \in C[0, 1]$. Let $f(x) = 1$ and $g(x) = x$ for $x \in [0, 1]$. Then $\|f\| = \sup |1| = 1$ and $\|g\| = \sup |x| = 1$. $(f + g)(x) = 1 + x \implies \|f + g\| = 2$. $(f - g)(x) = 1 - x \implies \|f - g\| = 1$. Checking the Parallelogram Law:

$$\|f + g\|^2 + \|f - g\|^2 = 2^2 + 1^2 = 5$$

$$2(\|f\|^2 + \|g\|^2) = 2(1^2 + 1^2) = 4$$

Since $5 \neq 4$, the Parallelogram law does not hold. Thus, the norm cannot be induced by an inner product. $\square$

2.1 Continuity of Inner Product

Lemma 2.1. Let X be an inner product space. Suppose that $\{x_n\}$ and $\{y_n\}$ are convergent sequences in X , with $x_n \rightarrow x$ and $y_n \rightarrow y$. Then:

$$\lim_{n \rightarrow \infty} \langle x_n, y_n \rangle = \langle x, y \rangle$$

Proof.

$$\begin{aligned} |\langle x_n, y_n \rangle - \langle x, y \rangle| &= |\langle x_n, y_n \rangle - \langle x_n, y \rangle + \langle x_n, y \rangle - \langle x, y \rangle| \\ &\leq |\langle x_n, y_n - y \rangle| + |\langle x_n - x, y \rangle| \\ &\leq \|x_n\| \|y_n - y\| + \|x_n - x\| \|y\| \quad (\text{Cauchy-Schwarz}) \end{aligned}$$

Since $\{x_n\}$ is convergent, it is bounded, so $\|x_n\| \leq C$. Also $\|y_n - y\| \rightarrow 0$ and $\|x_n - x\| \rightarrow 0$. Thus, $|\langle x_n, y_n \rangle - \langle x, y \rangle| \rightarrow 0$ as $n \rightarrow \infty$. $\square$

2.2 Orthogonality and Angles

By Cauchy-Schwarz, $|\langle x, y \rangle| \leq \|x\| \|y\|$. This implies $-1 \leq \frac{\text{Re}\langle x, y \rangle}{\|x\| \|y\|} \leq 1$ (provided $x, y \neq 0$). So the angle between x and y can be defined as $\theta = \cos^{-1} \left(\frac{\text{Re}\langle x, y \rangle}{\|x\| \|y\|} \right)$.

Definition 2.1. Let X be an inner product space. The vectors $x, y \in X$ are said to be **orthogonal** if $\langle x, y \rangle = 0$.

Remark 2.1. A set $\{e_1, e_2, \dots, e_k\} \subseteq X$ is said to be **orthonormal** if:

$$\langle e_n, e_m \rangle = \delta_{nm} = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{if } n \neq m \end{cases}$$

Lemma 2.2. (a) An orthonormal set $\{e_1, \dots, e_k\}$ in any inner product space X is linearly independent.

(b) In particular, if X is k -dimensional, then $\{e_1, \dots, e_k\}$ is a basis for X , and any vector $x \in X$ can be expressed as:

$$x = \sum_{n=1}^k \langle x, e_n \rangle e_n$$

Proof. (a) Suppose $\sum_{n=1}^k \alpha_n e_n = 0$. Taking the inner product with e_m :

$$0 = \left\langle \sum_{n=1}^k \alpha_n e_n, e_m \right\rangle = \sum_{n=1}^k \alpha_n \langle e_n, e_m \rangle = \alpha_m$$

Thus $\alpha_m = 0$ for all m . Hence the set is linearly independent.

(b) If $\{e_1, \dots, e_k\}$ is a basis, there exist scalars λ_n such that $x = \sum_{n=1}^k \lambda_n e_n$. Then $\langle x, e_m \rangle = \sum_{n=1}^k \lambda_n \langle e_n, e_m \rangle = \lambda_m$. $\square$

2.3 Gram-Schmidt Algorithm

Theorem 2.3. *Let $\{v_1, v_2, \dots, v_k\}$ be a linearly independent subset of an inner product space X , and let $S = \text{span}\{v_1, \dots, v_k\}$. Then there is an orthonormal basis $\{e_1, \dots, e_k\}$ for S .*

Proof. By induction on k . For $k = 1$, since $v_1 \neq 0$, take $e_1 = \frac{v_1}{\|v_1\|}$. Assume the result is true for k . Let $\{v_1, \dots, v_{k+1}\}$ be linearly independent. Let $\{e_1, \dots, e_k\}$ be the orthonormal basis for $\text{span}\{v_1, \dots, v_k\}$. Define:

$$b_{k+1} = v_{k+1} - \sum_{n=1}^k \langle v_{k+1}, e_n \rangle e_n$$

Note that $b_{k+1} \neq 0$ (otherwise v_{k+1} would be in the span of the previous vectors). Check orthogonality:

$$\langle b_{k+1}, e_m \rangle = \langle v_{k+1}, e_m \rangle - \sum_{n=1}^k \langle v_{k+1}, e_n \rangle \langle e_n, e_m \rangle = \langle v_{k+1}, e_m \rangle - \langle v_{k+1}, e_m \rangle = 0$$

Set $e_{k+1} = \frac{b_{k+1}}{\|b_{k+1}\|}$. Then $\{e_1, \dots, e_{k+1}\}$ is an orthonormal set with the same span. $\square$

2.4 Pythagorean Theorem and Hilbert Spaces

Theorem 2.4. *Let X be a k -dimensional inner product space and let $\{e_1, \dots, e_k\}$ be an orthonormal basis. Then for any $\alpha_n \in \mathbb{F}$:*

$$\left\| \sum_{n=1}^k \alpha_n e_n \right\|^2 = \sum_{n=1}^k |\alpha_n|^2$$

Definition 2.2 (Hilbert Space). An inner product space which is complete with respect to the metric associated with the norm induced by the inner product is called a **Hilbert Space**.

Example 2.2. 1. Every finite-dimensional inner product space is a Hilbert space.

2. ℓ^2 is a Hilbert space.

Lemma 2.5. *If H is a Hilbert space and $Y \subseteq H$ is a linear subspace, then Y is a Hilbert space iff Y is closed in H .*

3 Lecture 19: Orthogonal Complements

Definition 3.1. Let X be an inner product space and let A be a subset of X . The **orthogonal complement** of A is the set:

$$A^\perp = \{x \in X : \langle x, a \rangle = 0 \quad \forall a \in A\}$$

Example 3.1. If $X = \mathbb{R}^3$ and $A = \{(a_1, a_2, 0) : a_1, a_2 \in \mathbb{R}\}$, then $A^\perp = \{(0, 0, x_3) : x_3 \in \mathbb{R}\}$.

Lemma 3.1. Let X be an inner product space and $A \subset X$. Then:

- (a) $0 \in A^\perp$.
- (b) If $0 \in A$, then $A \cap A^\perp = \{0\}$. Otherwise $A \cap A^\perp = \emptyset$.
- (c) $\{0\}^\perp = X$ and $X^\perp = \{0\}$.
- (d) If A contains an open ball $B_r(a)$ for some $a \in X, r > 0$, then $A^\perp = \{0\}$. In particular, if A is a non-empty open set, $A^\perp = \{0\}$.
- (e) If $B \subseteq A$, then $A^\perp \subseteq B^\perp$.
- (f) $A^\perp$ is a closed linear subspace of X .
- (g) $A \subseteq (A^\perp)^\perp$.

Proof. (a) $\langle 0, a \rangle = 0 \implies 0 \in A^\perp$. (b) Suppose $x \in A \cap A^\perp$. Then $\langle x, x \rangle = 0 \implies x = 0$. (c) If $A = \{0\}$, $\langle x, 0 \rangle = 0$ for all x , so $X \subseteq A^\perp$. (d) Suppose $y \in A^\perp$ is non-zero. Let $a \in A$ be the center of the ball. Consider $a + \frac{r}{2\|y\|}y \in A$. Inner product with y must be 0: $\langle y, a \rangle + \frac{r}{2\|y\|}\langle y, y \rangle = 0 \implies 0 + \frac{r}{2}\|y\| = 0 \implies y = 0$. (f) Let $y, z \in A^\perp$. $\langle \alpha y + \beta z, a \rangle = \alpha \langle y, a \rangle + \beta \langle z, a \rangle = 0$. Thus it is a subspace. Let $x_n \in A^\perp$ and $x_n \rightarrow x$. Then $\langle x, a \rangle = \lim \langle x_n, a \rangle = 0$, so $x \in A^\perp$. Thus closed. (g) Let $a \in A$. For all $x \in A^\perp$, $\langle a, x \rangle = \overline{\langle x, a \rangle} = 0$. Thus $a \in (A^\perp)^\perp$. $\square$

3.1 Orthonormal Bases in Infinite Dimensions

Proposition 3.2. Any infinite-dimensional inner product space X contains an orthonormal sequence.

Question: For a general orthonormal sequence $\{e_n\}$ and $x \in X$, consider $x = \sum_{n=1}^{\infty} \langle x, e_n \rangle e_n$. (a) Does the series converge? (b) Does it converge to x ?

Lemma 3.3 (Bessel's Inequality - Preparation). Let $\{e_n\}$ be an orthonormal sequence. For any $x \in X$, the series $\sum |\langle x, e_n \rangle|^2$ converges and

$$\sum_{n=1}^{\infty} |\langle x, e_n \rangle|^2 \leq \|x\|^2$$

Proof. Let $y_k = \sum_{n=1}^k \langle x, e_n \rangle e_n$. Then:

$$\begin{aligned} \|x - y_k\|^2 &= \langle x - y_k, x - y_k \rangle \\ &= \|x\|^2 - \langle x, y_k \rangle - \langle y_k, x \rangle + \langle y_k, y_k \rangle \\ &= \|x\|^2 - \sum_{n=1}^k |\langle x, e_n \rangle|^2 \end{aligned}$$

Since $\|x - y_k\|^2 \geq 0$, we have $\sum_{n=1}^k |\langle x, e_n \rangle|^2 \leq \|x\|^2$. This holds for all k , so the series converges and is bounded by $\|x\|^2$. $\square$

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Theorem 4.1. Let H be a Hilbert space and $\{e_n\}$ an orthonormal sequence. Let $\{\alpha_n\}$ be a sequence in $\mathbb{F}$. The series $\sum_{n=1}^{\infty} \alpha_n e_n$ is convergent iff $\sum_{n=1}^{\infty} |\alpha_n|^2 < \infty$. If this holds, $\|\sum \alpha_n e_n\|^2 = \sum |\alpha_n|^2$.

Proof. Let $x_k = \sum_{n=1}^k \alpha_n e_n$. Then $\|x_k - x_j\|^2 = \sum_{n=j+1}^k |\alpha_n|^2$. Since $\sum |\alpha_n|^2$ converges, the tail goes to 0, so $\{x_k\}$ is Cauchy. Since H is complete, it converges to some x . Continuity of norm gives the equality. $\square$

Corollary 4.2 (Bessel's Inequality). Let H be a Hilbert Space and $\{e_n\}$ an orthonormal sequence. For any $x \in H$, the series

$$\sum_{n=1}^{\infty} \langle x, e_n \rangle e_n$$

converges.

Example 4.1. In $\mathbb{R}^3$, let $\{e_1, e_2\}$ where $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$. Let $x = (3, 0, 4)$. $\sum \langle x, e_n \rangle e_n = 3e_1 + 0e_2 = (3, 0, 0) \neq x$. This set does not span $\mathbb{R}^3$.

4.1 Characterization of Orthonormal Bases

Theorem 4.3. Let H be a Hilbert space and $\{e_n\}$ be an orthonormal sequence. The following are equivalent:

- (a) $\{e_n : n \in \mathbb{N}\}^\perp = \{0\}$.
- (b) $\overline{\text{span}}\{e_n : n \in \mathbb{N}\} = H$.
- (c) $\|x\|^2 = \sum_{n=1}^{\infty} |\langle x, e_n \rangle|^2$ for all $x \in H$ (Parseval's Identity).
- (d) $x = \sum_{n=1}^{\infty} \langle x, e_n \rangle e_n$ for all $x \in H$.

Proof. (a) $\Rightarrow$ (d): Let $y = x - \sum_{n=1}^{\infty} \langle x, e_n \rangle e_n$. Check inner product with e_m :

$$\langle y, e_m \rangle = \langle x, e_m \rangle - \sum_{n=1}^{\infty} \langle x, e_n \rangle \langle e_n, e_m \rangle = \langle x, e_m \rangle - \langle x, e_m \rangle = 0$$

Thus $y \in \{e_n\}^\perp$. By (a), $y = 0$, so $x = \sum \langle x, e_n \rangle e_n$.

(d) $\Rightarrow$ (b): Since partial sums are in the span, x is the limit of vectors in the span.

(d) $\Rightarrow$ (c): By continuity of norm and the previous theorem.

(b) $\Rightarrow$ (a): Suppose $y \in \{e_n\}^\perp$. Then $y \perp \text{span}\{e_n\}$. By continuity, $y \perp \overline{\text{span}}\{e_n\} = H$. So $\langle y, y \rangle = 0 \Rightarrow y = 0$. $\square$

Definition 4.1. An orthonormal sequence $\{e_n\}$ is called an **orthonormal basis** if any of the conditions in the previous theorem hold.

Remark 4.1. A Hilbert space has an orthonormal basis iff it is **separable** (contains a countable dense subset).

Theorem 4.4. (a) *Finite dimensional normed vector spaces are separable.*

(b) *An infinite-dimensional Hilbert space is separable iff it has an orthonormal basis.*

Proof. (a) Let X be finite-dimensional with basis $\{e_1, \dots, e_k\}$. The set of vectors $\sum q_n e_n$ with $q_n \in \mathbb{Q}$ is countable and dense. (b) We skip the proof. $\square$